

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 6: Standard Masses, Floor Limits & Cargo Restraints

Syllabus Source: CAE Oxford (pp. 107-124) | Pass Target: >= 90% AviationExam

1. EASA AIR-OPS STANDARD PASSENGER MASSES

Commercial operators may use EASA standard passenger masses instead of weighing every individual passenger at the gate:

Passenger Category	All Flights (>= 30 Passenger Seats)	Holiday Charter (>= 30 Seats)	Aircraft with 1 to 19 Seats
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Male Passenger	88 kg (including 6 kg hand luggage)	83 kg	96 kg (Male)
Female Passenger	70 kg (including 6 kg hand luggage)	69 kg	78 kg (Female)
All Adults (No Gender Split)	84 kg	76 kg	88 kg
Children (2 to 11 years)	35 kg	35 kg	35 kg
Infants (Under 2 years)	0 kg (if held on adult's lap)	0 kg	0 kg (Held on lap)

2. CARGO COMPARTMENT LIMITATIONS: FLOOR & RUNNING LOADS

Cargo holds are protected against structural collapse by two strict engineering limitations:

- Area Load Limit (Floor Intensity): Maximum allowable mass per unit surface area: kg/m^2 or lb/ft^2. Exceeding this punches through the floor panel! Formula: Load Intensity = Mass / Contact Area <= Floor Limit.
- Running Load Limit (Linear Load Limit): Maximum allowable mass per unit length of fuselage: kg/m or lb/in. Exceeding this breaks the transverse fuselage frames! Formula: Linear Load = Mass / Length <= Running Limit.
- Spreader Plates (Dunnage): If a dense, heavy box exceeds the floor load limit, placing rigid wooden or composite spreader plates beneath it increases the contact area, distributing the weight safely across multiple floor beams!

3. CARGO RESTRAINT & TIE-DOWN LOAD FACTORS

To prevent cargo breaking free during severe turbulence or emergency landing, lashings must restrain against design load factors (EASA CS-25): Forward: 9.0 g; Upward: 1.5 g; Sideward: 3.0 g; Aft: 1.5 g.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Floor Load Limits in Cargo Comp

Problem: A cargo box weighs 600 kg with a footprint of 1.2 m x 0.8 m. Compartment floor load limit is 500 kg/m². Can it be loaded safely? Step 1: Footprint Area = 1.2 m x 0.8 m = 0.96 m². Step 2: Applied Floor Load = Mass / Area = 600 kg / 0.96 m² = 625 kg/m². Step 3: Comparison: 625 kg/m² EXCEEDS the 500 kg/m² limit! Risk of floor structural damage! Step 4: Solution: Place on load spreader board with minimum area = 600 kg / 500 kg/m² = 1.20 m².

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Checking Floor Load Limits & Spreader Sizing**

SCENARIO (Classic EASA Calculation Drill): A cargo compartment has the following certified structural limits: - Maximum Floor Load Limit: 600 kg/m² - Maximum Running Load Limit: 400 kg/m A piece of industrial machinery has a mass of 1,200 kg. Its base dimensions are: Length = 2.0 meters, Width = 0.8 meters. QUESTION: Does this cargo exceed the Floor Load Limit or the Running Load Limit? If it exceeds, what is the MINIMUM contact area required using a spreader plate? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check the Running Load Limit: - Actual Linear Load = Mass / Length = 1,200 kg / 2.0 m = 600 kg/m. - Certified Running Limit = 400 kg/m. - VERDICT: 600 kg/m > 400 kg/m -> EXCEEDS RUNNING LOAD LIMIT by 200 kg/m! Step 2: Check the Floor Load (Area) Limit: - Contact Area = Length x Width = 2.0 m x 0.8 m = 1.6 m². - Actual Area Load Intensity = Mass / Area = 1,200 kg / 1.6 m² = 750 kg/m². - Certified Floor Limit = 600 kg/m². - VERDICT: 750 kg/m² > 600 kg/m² -> EXCEEDS FLOOR LOAD LIMIT by 150 kg/m²! Step 3: Calculate Minimum Spreader Plate Area required: - Area_min = Mass / Max Floor Limit = 1,200 kg / 600 kg/m² = 2.0 m². FINAL ANSWER: Both limits are exceeded! A spreader plate of AT LEAST 2.0 m² contact area and at least 3.0 m length (1,200/400) is mandatory.

[!] EXAM TRAP (AVIATIONEXAM) - Infant Mass & Hand Luggage Rules

- Infant mass (under 2 years) is 0 kg ONLY if held on an adult's lap. If occupying an individual seat, child mass (35 kg) must be applied! - Hand luggage (6 kg) is ALREADY INCLUDED in the standard passenger mass (88 kg male = 82 kg person + 6 kg bag). Do not add 6 kg twice!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA AIR-OPS on an aircraft with 30 or more passenger seats, what is the standard mass for an adult

- [A]: 84 kg
- [B]: 88 kg (including 6 kg hand luggage allowance)
- [C]: 70 kg
- [D]: 96 kg

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The EASA standard male passenger mass for aircraft with 30+ seats is 88 kg (82 kg body mass + 6 kg hand luggage).

Q2: What is the primary operational purpose of placing 'SPREADER PLATES' (dunnage) under dense cargo in

- [A]: To increase aircraft payload

- [B]: To increase the effective contact surface area, reducing floor load intensity (kg/m^2) below the certified structural floor limit
- [C]: To prevent fire
- [D]: To move the CG forward

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Spreader plates distribute the concentrated downward weight of heavy, compact cargo across a wider surface area, ensuring local pressure does not exceed the floor structural limit.

Q3: Under CS-25 regulations, what is the forward ultimate inertia load factor that cargo tie-downs and restraint

- [A]: 1.5 g
- [B]: 3.0 g
- [C]: 9.0 g forward
- [D]: 20.0 g

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. To protect occupants in an emergency landing, cargo restraints must hold cargo in place against a 9.0 g forward deceleration load factor.